

---

# ANALYZING LLM HALLUCINATION MITIGATION METHODS FOR LOW-RESOURCE LANGUAGES

**Pushpesh Thakur, Shravani Bhujbal & Srimedha Thummala**

Department of Computer Science

University of Washington

Seattle, WA, USA

## ABSTRACT

This study analyzes the effectiveness of various hallucination mitigation techniques in the generation of multilingual long-form text. Using the Multi-FACT evaluation framework (Shafayat et al. (2024)), the final fact evaluation score is derived from the FACTScore metric, which is the percentage of claims in the LLM text that are verified using Wikipedia retrievals. This study evaluates the factuality of the biographies of national leaders produced by a Qwen 8B model. Our analysis performs each hallucination mitigation strategy on high, medium, and low resource languages, specifically English, Hindi, Bengali, and Swahili. We compare a vanilla baseline against two distinct mitigation strategies: Retrieval-Augmented Generation (RAG) and Chain-of-Thought (CoT) prompting. Within RAG, we test two separate sub-strategies: one retrieves information from English Wikipedia while the other retrieves information from the target language Wikipedia. Similarly, for Chain-of-Thought prompting, we test two separate styles: one being Zero-Shot Chain-of-Thought and the other being Plan-and-Solve Chain-of-Thought. The findings of this study aim to clarify the relationship between the availability of linguistic resources and the effectiveness of reasoning-based versus retrieval-based strategies in reducing hallucinations.

## 1 INTRODUCTION

LLM use is rapidly becoming a critical tool across different domains like healthcare, law, and education. This has raised concerns about chatbots’ tendencies to produce hallucinations (convincing outputs that are presented as factual but are actually incorrect or unverifiable). This issue is even more perplexing in multilingual contexts, since there is a lack of language data and cultural context for many languages (referred to as low-resource languages). Users are starting to rely more on LLMs as reliable sources of information, so hallucinations pose risks of misinformation and poor decision-making. Addressing this problem is important to ensure that LLMs can be safely used in the real world.

In this paper, we test the effectiveness of different mitigation techniques, including Retrieval-Augmented Generation (RAG) and Chain-of-Thought (CoT) prompting, on various resource-level languages. We identify the following research question:

To what extent do retrieval and prompting strategies mitigate factual hallucinations in multilingual LLMs across varying language resource tiers?

To support our analysis, we integrated the Multi-FACT pipeline to obtain FACTScores (Min et al. (2023)) for biographies of political figures generated to evaluate hallucinations in different languages. Our analysis revealed that retrieval-augmented generation consistently improved FACTScores across all languages, with crosslingual RAG proving most effective for mid-resource languages (Hindi and Bengali) by utilizing generation in English Wikipedia’s denser knowledge base. Chain-of-Thought prompting, however, showed an inverse relationship with resource level: improving factuality for lower-resource languages (Swahili) while slightly degrading performance for higher-resource ones (English and Hindi). Notably, no single mitigation strategy universally

---

eliminated hallucinations across all languages, suggesting that hallucination mitigation techniques should be selected based on a language’s resource level rather than applied uniformly.

Our contributions are as follows:

- We replicated and extended the Multi-FAct pipeline to calculate the factual precision of biographies generated under aforementioned hallucination mitigation techniques by Qwen-8B across languages of different resource levels: English, Hindi, Bengali, and Swahili.
- We evaluated four mitigation strategies for improving factuality in multilingual generation. This includes English RAG, Native RAG, Zero-Shot CoT prompting, and Plan-and-Solve CoT prompting.
- We demonstrated the language-specific effects of different mitigation strategies, giving insights into the effectiveness of these techniques for different language resource levels.

## 2 RELATED WORK

**Multilingual LLM Factuality Evaluation** The Multi-FAct research paper (Shafayat et al. (2024)) dives deep into the limitations of factual LLM output for multilingual models, not just English. Multilingual LLMs lack transparency on factuality evaluations, especially for long-form generation outputs. The research paper introduces a simple pipeline to utilize FActScore to determine the reliability of various multilingual LLMs and regarding topics of regional diversity.

**Hallucination Mitigation Strategies** The blog (Sidhpurwala (2024)) gives a brief overview of what a hallucination is and four key methods of mitigating hallucinations, including fine-tuning models with domain-specific knowledge, implementing RAG, using advanced prompting techniques (such as Chain-of-Thought prompting or Few-Shot prompting), and utilizing guardrails. The blog is tailored for beginner-level readers with some key information and tools to implement mitigation techniques.

## 3 OUR PIPELINE

The overall method of this study is to assess the hallucinations in biographies of select world leaders, generated with and without different hallucination mitigation techniques, in different languages. The Multi-FAct pipeline was integrated as the primary measure of “hallucination” in multilingual generation with smaller open-source models. Qwen-8B, the model under assessment, is used to generate biographies in all four chosen languages. As part of the Multi-FAct pipeline, the biographies of non-English languages are translated to English using GPT-4. This ultimately follows the same reasoning as the Multi-FAct pipeline: making fact verification in English will let us leverage the larger data store that is English Wikipedia.

The FActScore evaluation metric is set-up to retrieval + Mistral. This set-up uses Mistral-7B for breaking a biography into atomic facts, where each atomic fact is independently verifiable with the FActScore’s Wiki database. It then retrieves information from the Wiki database to verify these facts. The final output is a FActScore for the generated biographies, which is a simple ratio of number of atomic facts verified by the the database to the total number of atomic facts generated. A higher FActScore implies higher factuality, and thus lower hallucination.

The languages included in this study were chosen based on the varying resource level tiers. We classified the languages in three tiers with English being high-resource, Hindi and Bengali being medium-resource, and Swahili being a low-resource language. These language classifications are based on the paper “The State and Fate of Linguistic Diversity and Inclusion in the NLP World” by Joshi et al. (2020). The world leaders included in the study are Xi Jinping, Barack Obama, and Dilma Rousseff. These figures were carefully chosen to ensure no one language has a geographic/native advantage in terms of the resources available for any leader. For example, choosing Shaikh Hasina would’ve given Bengali an unfair advantage over other languages in terms of the resources available since Shaikh Hasina is a Bangladeshi figure where Bengali is the native language.



Figure 1: Language hierarchy based on resources available.

Each section of this paper re-generates the biographies for the same three world leaders but with different mitigation strategies implemented. This allows us to evaluate the different mitigation strategies on each of the languages of the four languages independently and compare them with the baseline. The following figure provides an overview of our procedure:

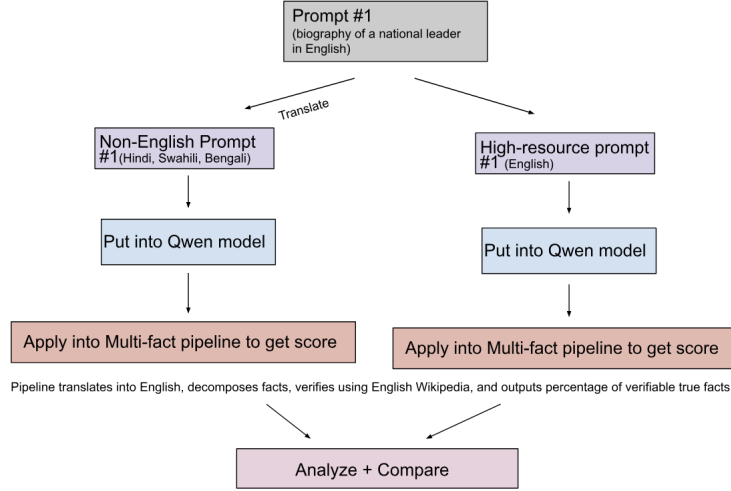


Figure 2: The process used in the experiment. 1) Create a prompt to generate a biography of a national leader using a mitigation strategy in English. 2) Translate prompt into the designated language. 3) Put new prompt into Qwen model and capture output. 4) Put generated output into Multi-FACT pipeline. 5) Analyze and compare.

## 4 BASELINE

For the baseline, we prompted vanilla Qwen-8B to generate biographies without allocating tokens for thinking. All baseline prompts followed a simple format of:

“Write biography of [world leader name]”

We then translated the outputs (biographies) from the target language into English using GPT-4 as a part of the Multi-FACT pipeline. As shown in Table 2 in the Multi-FACT paper (Shafayat et al. (2024)), translation with GPT-4 does not significantly affect the FActScore obtained. Finally, we evaluated the translations to obtain the FActScores.

### 4.1 RESULTS + ANALYSIS

| Language | Xi Jinping | Barack Obama | Dilma Rousseff | AVG FActScore  |
|----------|------------|--------------|----------------|----------------|
| English  | 0.86893    | 0.90640      | 0.70879        | <b>0.82804</b> |
| Hindi    | 0.75281    | 0.83333      | 0.71831        | <b>0.76815</b> |
| Bengali  | 0.08165    | 0.60825      | 0.68750        | <b>0.45913</b> |
| Swahili  | 0.17073    | 0.22313      | 0.70833        | <b>0.36740</b> |

Table 1: FActScore Evaluation Results for Baseline Model.

The results generated for the world leaders showed the expected trend in FActScores. English, being a high-resource language, achieved the highest average (0.82804), followed by Hindi (0.76815), then Bengali (0.45913), and finally Swahili (0.36740). These results are consistent with the idea that FActScores increase with higher resource levels.

The Bengali biography notably underperformed for Xi Jinping, as it consistently generated a biography for Shimon Peres despite being explicitly prompted otherwise. This explains the anomalous FActScore of 0.08165. In addition, we saw that Swahili’s generation overperformed for Dilma Rousseff, as it repeatedly asserted the simple fact that the leader “did not have a wife named Andika”, which is objectively true but not a biography. With the exception of Swahili, each language performed best when generating biographies for Barack Obama, likely due to Qwen being trained on English-dominant training data.

## 5 MITIGATION STRATEGY #1: RETRIEVAL-AUGMENTED GENERATION (RAG)

We implemented retrieval in each language. We injected the first  $n$  lines of the English Wikipedia page for the respective person we are generating the biography for. The formatting for the RAG prompt is:

“Using the following factual context:  
[N lines of wiki]  
Write a biography for [Figure].”

In addition, for non-English languages, we also generate another prompt that involves retrieving the information from the designated language’s native Wikipedia page. See section 5.2.

We collected results using the following two sub-methods of RAG prompts: Crosslingual RAG and Monolingual RAG.

### 5.1 CROSSLINGUAL RAG

Crosslingual RAG, as a hallucination mitigation strategy, aims to bridge the resource gap in high-resource and low-resource languages by simply using content from a high-resource language (English) to aid the generation in lower-resource languages. We retrieve lines from each leader’s respective English Wikipedia, and the biographies are produced in the target language. We inject these retrieved lines into the prompt before generating the biography in each language.

#### 5.1.1 RESULTS

Our results for Crosslingual RAG are as follows:

| Language | Xi Jinping | Barack Obama | Dilma Rousseff | AVG FActScore  |
|----------|------------|--------------|----------------|----------------|
| English  | 0.92248    | 0.99074      | 0.91453        | <b>0.94258</b> |
| Hindi    | 0.94915    | 0.80822      | 0.82258        | <b>0.85998</b> |
| Bengali  | 0.88888    | 0.90000      | 0.91358        | <b>0.90082</b> |
| Swahili  | 0.50000    | 0.35294      | 0.59259        | <b>0.48184</b> |

Table 2: FActScore Evaluation Results for RAG w/ English Wiki

#### 5.1.2 ANALYSIS

As seen in the table, the crosslingual scores in all languages showed improvements compared to their own respective baseline scores. Moreover, crosslingual RAG improves the Hindi average FActScore above English average baseline (0.82804). Similarly, crosslingual Bengali RAG also achieves a higher FActScore than English baseline average. This shows how crosslingual RAG is effective

in bridging the resource-gap to a certain extent. It is important to note that Hindi and Bengali crosslingual was still not able to surpass English crosslingual RAG score (0.94258). Though Swahili crosslingual score is higher than Swahili baseline (0.48184), it was not able to surpass English baseline unlike other languages.

## 5.2 MONOLINGUAL RAG

For monolingual RAG, we retrieve lines from each leader’s respective target language Wikipedia page to aid the biography generation. We then inject these retrieved lines into the prompt before generating the biography in each language. For example, if we performed monolingual RAG for Xi Jinping in Bengali, we would insert information retrieved from Bengali Wikipedia’s page on Xi Jinping.

### 5.2.1 RESULTS

Our results for monolingual RAG are as follows:

| Language | Xi Jinping | Barack Obama | Dilma Rousseff   | AVG FActScore  |
|----------|------------|--------------|------------------|----------------|
| English  | 0.92248    | 0.99074      | 0.91453          | <b>0.94258</b> |
| Hindi    | 0.80435    | 0.83333      | 0.8              | <b>0.81256</b> |
| Bengali  | 0.82143    | 0.67272      | N/A - No article | <b>0.74701</b> |
| Swahili  | 0.64286    | 0.65217      | N/A - No article | <b>0.64752</b> |

Table 3: FActScore Evaluation Results for RAG w/ Native Wiki

Note: English will have the same scores in crosslingual and monolingual RAG as both retrieve information from the English Wikipedia. It is also important to note that articles on Dilma Rousseff did not exist in Bengali and Swahili, which further highlights the gap in resources between English and lower-resource languages.

### 5.2.2 ANALYSIS

The results of monolingual RAG support our hypothesis that the mitigation techniques that rely on information sources in target languages perform proportionally to each language’s resource level. Moreover, each language achieves a higher score using crosslingual RAG (with error correction in Swahili discussed in section 5.1.3) than monolingual RAG. This confirms that aiding generations in lower-resource languages with content in high-resource languages may help mitigate hallucinations to a certain extent mainly because it provides denser source of facts. However, it hinges on the generation model’s capability to translate English to the target language.

## 5.3 ERROR ANALYSIS

We noted that the output relies significantly on Qwen-8B’s ability to translate the injected content/context from English Wikipedia to the target language before producing the biographies. This introduces a new source of error, which was affecting the FActScore. To discover this, we re-ran the crosslingual biography generation with English Wikipedia, but this time the English content is pre-translated to the target language before inserting into the prompt. GPT-4 was the dedicated model for translating the English Wikipedia content to the target language. We decided to perform this on all non-English languages, but our main focus remained on Swahili. This was the main anomaly of our findings, since the Swahili crosslingual RAG (which uses English Wikipedia) performs worse than the Swahili monolingual RAG (which uses Swahili Wikipedia), which was not consistent with results for other non-English languages. Moreover, it does not make sense since English Wikipedia is a much denser knowledge source. This allowed us to isolate the translation and understand how significantly the FActScore changes when the English Wikipedia context is translated by GPT-4 vs when it is internally translated by Qwen itself.

The following results table shows the scores for direct context injection vs. pre-translated context injection:

| Language | FactScore w/o GPT pre-translation | FactScore w/ GPT pre-translation |
|----------|-----------------------------------|----------------------------------|
| Swahili  | 0.48184                           | 0.69955                          |
| Hindi    | 0.85998                           | 0.88455                          |
| Bengali  | 0.90082                           | 0.9132                           |

Table 4: Crosslingual RAG context translation comparison

The results above confirmed that a great source of error was in Qwen’s ability to translate from English to Swahili. The average score for Swahili biographies based on English Wikipedia content pre-translated with GPT showed a hike of almost 22%.

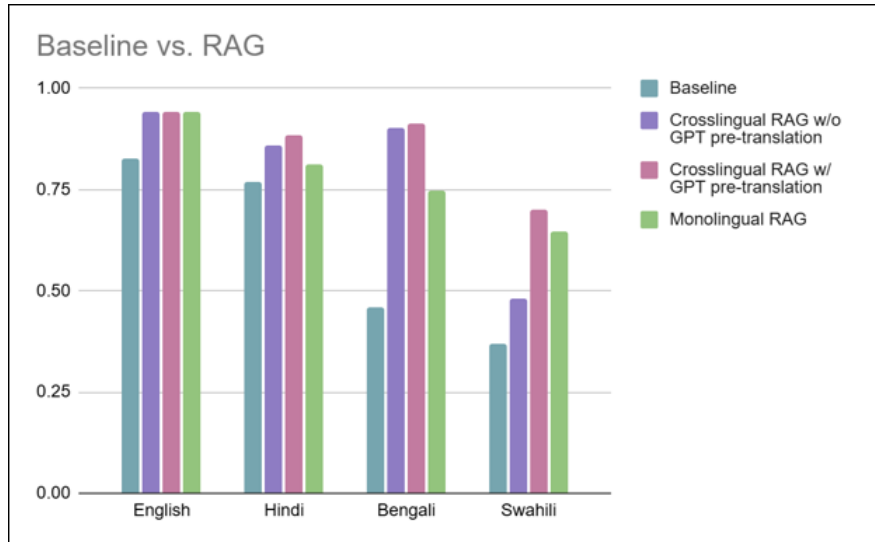
As noted before, we also ran the GPT context pre-translation method on Hindi and Bengali. However, the differences in the scores is only 3%.

#### 5.4 COMPARISON ANALYSIS

In Hindi and Bengali, we can see that the injection of information from English Wikipedia was more effective than retrieving from each language’s native Wikipedia. This is likely because English Wikipedia has a higher factual density.

However, in Swahili, we see the opposite happen. The native Wiki retrieval outperforms the English Wiki. This can potentially be attributed to Qwen’s translation capabilities discussed in section 5.1.3. When we inject English context into a Swahili prompt, the model needs to read the English and generate the context in Swahili for it to use. For Hindi/Bengali, Qwen may have seen enough of data in the languages during its training to be able to translate and bridge the gap decently. Perhaps for Swahili, that gap is wider. So when it translates from English to Swahili, it produces a contextual representation that is not what the actual context was, introducing more noise and hurting the generation rather than helping. Even though the native Swahili Wikipedia page is sparser, the loss of context through crosslingual transfer does not happen, so the facts it does provide can actually be used in generation, which may have resulted in higher score in monolingual RAG than crosslingual RAG in Swahili.

The following graph provides a comparison of scores of the RAG techniques with the baseline.



---

## 6 MITIGATION STRATEGY #2: CHAIN-OF-THOUGHT PROMPTING (CoT)

Chain-of-Thought prompting is a prompt engineering technique that focuses on guiding a model through logical problem-solving through a step-by-step reasoning process (Kavlakoglu et al. (2024)). This strategy allows the model to assign tokens for “thinking” along with a prompt to incentivize the model to think before it produces the biography.

Within CoT, we analyzed the FActScores of two versions: zero-shot prompting and plan-and-solve prompting. Note that we increased the maximum number of tokens allowed to use from 2048 to 4096 for both Hindi and Bengali. These biographies were getting truncated due to the model’s token expenditure from thinking and the languages being more expensive in terms of token use as they don’t use the same Latin script as English and Swahili (Tamang & Bora (2024)).

### 6.1 ZERO-SHOT PROMPTING

For both Chain-of-Thought style biography generations, the model was set to enable “thinking,” which allows it to expend tokens on thinking before it proceeds to generate the biographies.

The study included the following prompt for zero-shot Chain-of-Thought prompting:

“Question: Write a biography of [figure].  
Answer: Let’s think step by step.”

#### 6.1.1 RESULTS

The results for zero-shot CoT prompting are presented in the following table:

| Language | Xi Jinping | Barack Obama | Dilma Rousseff | AVG FActScore  |
|----------|------------|--------------|----------------|----------------|
| English  | 0.86036    | 0.90857      | 0.67157        | <b>0.81350</b> |
| Hindi    | 0.80672    | 0.79439      | 0.64462        | <b>0.74858</b> |
| Bengali  | 0.75701    | 0.85256      | 0.67133        | <b>0.7603</b>  |
| Swahili  | 0.58333    | 0.25         | 0.28571        | <b>0.37301</b> |

Table 5: Zero-Shot Prompting Results

#### 6.1.2 ANALYSIS

As seen by the data, English has the highest average FActScore, which is consistent with initial predictions given the amount of data English (as a high-resource language) has regarding each of the political figures. The FActScore is slightly less than the baseline experiment score average of 0.82804. Surprisingly, the scores for the remaining languages are relatively similar to the baseline results. More specifically, Hindi through CoT is slightly less (0.74858) than the average score for Hindi from the baseline results (0.76815). On the other hand, the score for Bengali (0.7603) is much higher than its baseline results of 0.45913. Swahili had a higher score (0.37301) than its average baseline score (0.36740).

The broader observation is that using zero-shot Chain-of-Thought prompting as a mitigation method reduces the score of high/mid-resource languages (English, Hindi) with an exception of Bengali while increasing the scores of the low-resource language (Swahili).

### 6.2 PLAN-AND-SOLVE PROMPTING

The study included the following prompt:

“Question: Write a biography of [figure].  
Answer: Let’s think step by step to ensure all dates and roles are 100% factual.  
1. Birth details:

- 
- 2. Education:
  - 3. Major Political Roles:
  - 4. Final Biography.”

### 6.2.1 RESULTS

The results for plan-and-solve CoT prompting are presented in the following table:

| Language | Xi Jinping | Barack Obama | Dilma Rousseff | AVG FActScore  |
|----------|------------|--------------|----------------|----------------|
| English  | 0.79592    | 0.90449      | 0.73786        | <b>0.81276</b> |
| Hindi    | 0.60629    | 0.78571      | 0.67521        | <b>0.68907</b> |
| Bengali  | 0.74038    | 0.66667      | 0.80246        | <b>0.73650</b> |
| Swahili  | 0.57627    | 0.71667      | 0.37333        | <b>0.55542</b> |

Table 6: Plan-and-Solve Prompting Results

### 6.2.2 ANALYSIS

As seen in the data, the FActScores for English was not significantly impacted by this mitigation technique (baseline was 0.82804). Hindi showed a decent decline in performance of around 8 points compared to its own baseline results. We saw a significant increase in performance for Bengali and Swahili when it came to plan-and-solve prompting, with Bengali improving from its baseline score of 0.45913 and Swahili improving from its baseline score of 0.36740.

## 6.3 GENERAL CHAIN-OF-THOUGHT ANALYSIS

In both methods, we saw similar results in that they performed worse than baseline in English and Hindi but improved from baseline for Bengali and Swahili. The reason for this could be that the baseline scores for low-resource languages are low due to language and grammar errors (such as looping, switching to English, losing grammar, etc.); however, with CoT incentive, the model generates more “step-by-step” tokens that stabilize the linguistic issues produced in the baseline output, leading to a higher FActScore. The decrease in FActScore in high-resource languages can be explained through introduction of logical noise (resulting in over computation) and/or a misconception by the model to be consistent with its answers (resulting in consistently incorrect answers after its first incorrect fact).

We also noticed that despite Hindi and Bengali being in the same resource tier, Hindi’s FActScore worsened as compared to its baseline while Bengali’s FActScore improved compared to its respective baseline. The reason for this may be that Hindi is still higher resource than Bengali even if they sit in the same tier (see Fig 1). Bengali’s resource level is high enough that the model can think and reason in Bengali, but low enough that the model lacks the ability to recall the Bengali information it was trained on. This makes CoT a perfect mechanism that leverages thinking to recall the information, improving the overall FActScore. However, Hindi’s resource-level may be past this point where the model is adapt to think as well as recall causing CoT to introduce noise due to overthinking, which may be resulting in the decrease of FActScore observed.

## 6.4 PLAN-AND-SOLVE VS. ZERO-SHOT ANALYSIS

Interestingly, plan-and-solve prompting performed worse than zero-shot. This seems counter-intuitive, since giving a model the exact structure should theoretically improve the factual precision because we prompted for the exact details we want. One reason for this could be that the structure is too rigid. By giving a structure, the model is pressured to fill in each slot with specific claims even when it is unsure. The freer prompt of zero-shot lets it glide over topics it is not confident on, either by giving vague claims or skipping the category entirely. Ultimately, more unsure claims likely means more incorrect facts, decreasing factual precision. Additionally, telling the model to be 100% factual may introduce overconfidence instead of helping factuality. The model could be more likely to be more likely to strongly assert incorrect facts rather than be transparent about low-confidence responses, which is exactly what FActScore penalizes.



We saw an exception with Swahili, where the plan-and-solve prompting greatly outperforms zero-shot. This is likely because, as a lower-resource language, the model’s thinking ability in Swahili is significantly worse than other languages, to the point where introducing structure gives a solid foundation for a biography rather than being a catalyst for overthinking.

## 7 CONCLUSION

In this paper, we examined the effectiveness of different hallucination mitigation methods across multiple language resource tiers. The general trend observed was that both RAG and CoT aid in hallucination mitigation for majority of the languages.

When applying retrieval-augmented generation, we saw that it was generally effective at improving FActScores across all languages. Crosslingual RAG demonstrated the strongest improvements, boosting Bengali and Hindi to above English baseline score. This attributes to the fact that mid-resource tiers have generation fluency, but often lack factual grounding, which is exactly what RAG helps with. On the other hand, trends in low-resource languages like Swahili show that crosslingual RAG is limited by the model’s ability to transfer content from English to the target language. This explains why monolingual RAG may perform better than crosslingual RAG in low-resource languages despite crosslingual RAG retrieving information from denser resources in English (or other high-resource languages).

Chain-of-Thought prompting showed lesser and more inconsistent results, improving lower-resource languages but degrading performance for higher-resource ones.

Ultimately, no single strategy was able to completely mitigate hallucinations across all tested languages. The effectiveness of each technique is tied to the amount of resources a language has. This suggests that hallucination mitigation techniques should be applied based on a language’s resource level rather than being uniformly applied to anything.

| Resource Level | Language | Crosslingual RAG | Monolingual RAG | Zero-Shot CoT | Plan-and-Solve CoT |
|----------------|----------|------------------|-----------------|---------------|--------------------|
| <b>High</b>    | English  | ✓                | ✓               | ✗             | ✗                  |
| <b>Mid</b>     | Hindi    | ✓                | ✓               | ✗             | ✗                  |
|                | Bengali  | ✓                | ✓               | ✓             | ✓                  |
| <b>Low</b>     | Swahili  | ✓                | ✓               | ✓             | ✓                  |

Table 7: Mitigation Strategy Effectiveness by Language Resource Level.  
 ✓ indicates improvement over baseline; ✗ indicates no improvement.

## 8 LIMITATIONS

One limitation came with our evaluation metric, FActScore. It is effective at measuring factual precision but is not a good judge of relevancy to the prompt. We saw this in Swahili’s baseline generation for Dilma Rousseff, where Qwen generated a response that was not an actual biography. It still produced atomic facts to score, so it received a higher FActScore (0.70833).

Additionally, due to limited time and computation resources, the scale of the experiments are smaller and can be expanded in future research. For one, we only ran the experiments on Qwen-8B instead of multiple, stronger multilingual models. Additionally, we could have better measured consistency by running experiments with more languages across resource levels and more leaders in terms of quantity and diversity.

---

## REFERENCES

- Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The State and Fate of Linguistic Diversity and Inclusion in the NLP World. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 6282–6293, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.560. <https://aclanthology.org/2020.acl-main.560.pdf>
- Eda Kavlakoglu, Vrunda Gadesha, and Vanna Winland. What is chain of thought (CoT) prompting? *IBM Think Topics*, 2024. URL <https://www.ibm.com/think/topics/chain-of-thoughts>.
- Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. URL <https://arxiv.org/abs/2305.14251>
- Sheikh Shafayat, Eunsu Kim, Juhyun Oh, and Alice Oh. Multi-FACT: Assessing Factuality of Multilingual LLMs using FActScore. *Proceedings of the Conference on Language Modeling (COLM)*, 2024. URL <https://arxiv.org/abs/2402.18045>.
- Huzaifa Sidhpurwala. When LLMs day dream: Hallucinations and how to prevent them. *Red Hat Blog Insights*, September 2024. URL <https://www.redhat.com/en/blog/when-llms-day-dream-hallucinations-how-prevent-them>.
- Sagar Tamang and Dr. Dibya Jyoti Bora. Evaluating Tokenizer Performance of Large Language Models Across Official Indian Languages. *arXiv preprint arXiv:2411.12240v2*, 2024. URL <https://arxiv.org/html/2411.12240v2>.

## APPENDIX

**GitHub:** <https://github.com/MLCapstone/multilingual-hallucination>

## AUTHOR CONTRIBUTIONS

**Pushpesh Thakur:** Contributed to research and project design, running experiments (generating biographies and evaluations for all languages and mitigation strategies), and conducting analyses and interpretations of the results that followed. Set up information retrieval pipeline for RAG biography generations and contributed to integrating Multi-FACT pipeline for FActScore evaluation. Contributed to report writing, including the final project paper.

**Shravani Bhujbal:** Contributed to the conceptualization of the project idea and deigned experiments methodologies. Configured the Qwen 8B model for biography generations leveraging Ollama, which was used across all experiments. Contributed to integrating Multi-FACT pipeline to use FActScore evaluation. Other contributions include running experiments and interpreting and analyzing results.

**Srimedha Thummala:** Contributed to documentation by creating baseline result visualization, creating presentation, supporting report writing, and writing final report. Within the experiment, explored hallucination mitigation to implement in the project, executed Chain-of-Thought experiments, and performed analyses of results.